Lab 1: Look at the data

Today we started on a new area, and that can feel confusing at first. This lab has two parts, and **you choose**.

Part 1: Practise repeats every step we did today, with other columns and other questions. The principle is the same. Do this part if you want to try all the steps once more and feel sure about them.

Part 2: Go deeper is for you who felt "I understood this, and I want to go further". The questions are more open, and you combine the tools from today.

I do not expect you to do both parts. It is your own choice. You are also welcome to contact me if you want to talk about which part to take.

You work in your own notebook, with the same file as today: `data/listings.csv`. Call the table `listings`, and write a short comment above each answer, in your own words. If you get stuck: read the error message from the bottom up, print the thing you are working with, and look at the lecture notebook. Then ask.

Part 1: Practise

Repeat the steps from today with other columns. Same principle, new questions.

Tools you need

```
import pandas as pd
listings = pd.read_csv('data/listings.csv')        # load the file
listings.shape      listings.head()      listings.info()
listings[['a', 'b']].count()                       # values that are NOT missing
listings[['a', 'b']]                               # pick columns (double brackets)
listings['x'].describe()      listings['x'].nunique()
mask = listings['x'] >= 2     mask.sum()     mask.mean()
listings[(mask1) & (mask2)]                        # & means AND
listings[cols].sort_values('x', ascending=False).head(5)
```

Tasks

1 **Load the file and take a first look**
Create a new notebook called `lab1.ipynb` next to your `data` folder. Import pandas, load the file into `listings`, and print the shape. Then run `head()` and `info()`. How many rows and columns are there?

2 **Missing values in two columns**
Run `listings[['bedrooms', 'beds']].count()`. The result is the number of values that are **not** missing. Which of the two columns has more missing values, and how many? Use the number of rows from task 1.

3 **Choose columns**
Make a list called `view` with the columns `name`, `neighbourhood_cleansed`, `bedrooms`, `beds` and `price`. Show the first five rows of `listings[view]`.

On purpose: Then run `listings['name', 'price']`. Read the error. What do you have to change?

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4 Describe the numbers

Use `describe()` on the columns `bedrooms`, `beds`, `availability_365` and `minimum_nights`, rounded to two decimals. Answer in a comment: which column has the biggest maximum? How many nights is the biggest `minimum_nights`, and does it seem realistic for a rental?

5 Describe the text

Use `describe()` on `property_type` and `neighbourhood_cleansed`. How many different values does each column have, and which value is the most common? Check the number for `property_type` with `nunique()`.

6 The price is text

Print the first three values of `price` and the dtype of the column. Then run the line below. The top row has the price `$999.60`, but you have already seen a listing that costs `$1,091.00`. Why is `$999.60` first? Explain in a comment, in your own words. We fix the price on Monday 16 November.

```
listings[['name', 'price']].sort_values('price', ascending=False).head(3)
```

7 Filter rows

(a) Make a mask for listings with two or more bedrooms. Use `sum()` and `mean()` on it: how many are there, and what share of all listings?
(b) Find the listings that are in `Östermalms`, are `Entire home/apt` and have two or more bedrooms. Start with `and` between the conditions and read the error. Then fix it with `&` and parentheses, and print the shape.

On purpose: Use `and` first, and read the error from the bottom up.

8 Sort and pick

Find the five listings in `Östermalms` with the most reviews. Show `name`, `room_type` and `number_of_reviews`, with the biggest number of reviews first. Which room type does the listing at the top have?

Expected

Task 1: (4574, 84). **Task 2:** 4572 and 4473 values, so `beds` is missing 101 (and `bedrooms` 2).
Task 3: `listings[['name', 'price']]` with double brackets. The first row is "Double room in central Stockholm with Wi-Fi", $1,091.00.
Task 4: count 4572 / 4473 / 4574 / 4570 and max 12 / 22 / 365 / 500. **Task 5:** 44 property types and 14 neighbourhoods. The most common are Entire rental unit (2442) and Södermalms (1145).
Task 6: the first three prices are '$1,091.00', '$702.00', '$999.00' and the dtype is str. The top row is $999.60. **Task 7:** (a) 1825 listings, share about 0.399. (b) (93, 84).
Task 8: the top row is "Great Room in City Center" with 855 reviews. The next four have 410, 344, 326 and 312.

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Part 2: Go deeper

Here you combine the tools from today to answer more open questions. There is no new syntax, but you have to think about what the numbers mean.

Tools you need

```
central = ['Södermalms', 'Östermalms', 'Norrmalms', 'Kungsholmens']
listings['x'].isin(central)       ~mask         mask1 & mask2
listings[mask]['room_type'] == 'Private room'     # a mask inside a mask
listings.sort_values(['a', 'b'], ascending=[False, False])
listings.nlargest(5, 'x')
table.loc[347, 'name']      table.iloc[3]['name']      table.reset_index(drop=True)
```

Tasks

1 **Central or not**

Make a list `central` with the four central neighbourhoods `Södermalms`, `Östermalms`, `Norrmalms` and `Kungsholmens`, and make a mask with `isin()`. How many listings are central, and what share is that of all listings?

Then compare the **share of private rooms** among the central listings with the share outside the centre (use ~). Is it higher in the centre or outside?

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2 **Room for a group**

What share of the `Entire home/apt` listings have room for four guests or more (`accommodates`)? And what share of all the other listings? Write one sentence in a comment about what the difference tells you, and one about what it does not tell you.

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3 **Best reviewed, fairly**

A listing with one review that says 5.0 is not as impressive as a listing with 100 reviews that says 5.0. So first keep only listings with **at least 50 reviews**. How many of them have the rating exactly 5.0?

Then show `name`, `neighbourhood_cleansed`, `number_of_reviews` and `review_scores_rating` for the top five, in two ways: with `sort_values` on rating **and** then reviews (both biggest first), and with `nlargest(5, 'review_scores_rating')`. The lists are different. Why? Explain in a comment.

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4 **Label or position**

Take the sorted top five from task 3 and save it in a variable `top`. Print the name of the listing with the label `347` with `loc`, and of the fourth row with `iloc[3]`. They are the same listing. Then run `top = top.reset_index(drop=True)` and try `top.loc[347, 'name']` again. What happens, and why?

On purpose: The last line should give an error. Read it from the bottom up.

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5 What does a zero mean?

(a) How many listings have `bedrooms` equal to 0? Which `room_type` do most of them have? What could 0 bedrooms mean in reality?

(b) How many listings have `availability_365` equal to 0? Two explanations are possible: the listing is fully booked for the whole year, or it is closed and cannot be booked. Check how many of these listings have `number_of_reviews` equal to 0. Does that help you choose one explanation? Write in a comment what you **can** and **cannot** conclude.

6 Your own question

Come up with **one question** about the data that you can answer with the tools from today. Write the question as a comment, answer it with code, and finish with a comment about what the answer does **not** tell you. That last part is the most important one.

Expected

Task 1: 2331 central listings, share about 0.510. The share of private rooms is about 0.151 in the centre and about 0.194 outside it.
Task 2: about 0.557 of the entire homes and about 0.093 of the other listings.
Task 3: 14 listings. The two top fives are different, because many listings have exactly 5.0, so the order depends on how ties are handled. The first with `sort_values` is "Breakfast paradise City Stockholm" (159 reviews).
Task 4: both lines give the listing called "Stockholm". After `reset_index`, `top.loc[347, 'name']` gives a KeyError: the label 347 does not exist anymore, the labels are 0 to 4.
Task 5: (a) 257 listings, 231 of them `Entire home/apt`. (b) 1494 listings, and 231 of them have no reviews.

Stuck? Read the error message from the bottom up, print the data you are working with, and then ask.